

Watch: Tedx talk by Grant Sanderson

"What makes people engage with math?"

n -dim vector space $\mathbb{R}^n = \left\{ \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \mid v_i \in \mathbb{R} \right\}$

• operations on vectors: can add, subtract vectors of same size
can scale a vector by a real number

• matrices: can add, subtract matrices of same size
can scale a matrix by a real number

• can multiply matrices of appropriate size

$$\left. \begin{array}{l} A \quad p \times q \\ B \quad q \times r \end{array} \right\} AB \text{ is a } p \times r \text{ matrix}$$

• For matrices of appropriate sizes

$$\left. \begin{array}{l} (AB)C = A(BC) \\ (A+B)C = AC + BC \\ D(A+B) = DA + DB \end{array} \right\} \text{Check!}$$

sketch:
$$\begin{aligned} \text{LHS} &= [(AB)C]_{ij} = \sum_k (AB)_{ik} C_{kj} \\ &= \sum_k \left[\sum_l (A_{il} B_{lk}) \right] C_{kj} \end{aligned}$$

$$\begin{aligned} \text{RHS} &= [A(BC)]_{ij} = \sum_x A_{ix} (BC)_{xj} \\ &= \sum_x A_{ix} \left[\sum_y B_{xy} C_{yj} \right] \end{aligned}$$

LHS = RHS b/c of $\left\{ \begin{array}{l} \text{distributivity \& associativity for } \mathbb{R}, \\ \text{reindexing } x \leftrightarrow l, \text{ rearranging} \\ y \leftrightarrow k, \text{ summation} \end{array} \right.$

- With distributivity & scaling we can express matrix multiplication using only matrix units

E_{ij} = matrix with a single nonzero entry 1 in i th row & j th column

$$(a_{ij}) = \sum_{ij} a_{ij} E_{ij} \quad \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} = aE_{11} + bE_{12} + \dots + fE_{23}$$

$$E_{ij} E_{kl} = \begin{cases} E_{il} & \text{if } j=k \\ 0 & \text{if } j \neq k \end{cases}$$

$$= \delta_{jk} E_{il} \quad \delta_{jk} = \begin{cases} 1 & j=k \\ 0 & j \neq k \end{cases}$$

"Kronecker delta"

Rmk: We can do linear algebra by taking scalars from any "field" (~~world~~^{Set} with 4 arithmetic operations satisfying "usual" properties) e.g. instead of $\mathbb{R}$ we can use $\mathbb{Q}$, $\mathbb{C}$, "integers modulo p " (p is prime)

Why not define multiplication of two vectors in $\mathbb{R}^3$ in "natural way":

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} \cdot \begin{bmatrix} d \\ e \\ f \end{bmatrix} := \begin{bmatrix} ad \\ be \\ cf \end{bmatrix}$$

Such defⁿ would give

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \uparrow \cdot \rightarrow = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

[So NOT interesting]

(not compatible with geometric understanding of vectors) $\nwarrow \nearrow \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Back to systems of linear equations

$$Ax = b \quad A \text{ } r \times c \text{ matrix, } x \in \mathbb{R}^c, b \in \mathbb{R}^r$$

unknown vector

Systematic procedure to solve $Ax=b$ row reduction

Ingredients

- scaling an equation by a nonzero scalar
- adding a multiple of one equation to another
- reorder equations

translate into similar operations on rows of the matrix $[A|b] \rightarrow$ augmented matrix associated to the system $Ax=b$

- HW
- Read about reduced echelon form of matrices
 - Visualize geometrically possibilities in solving one/two/three equations in three variables.
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